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Communication enabled circuit breakers and circuit breaker panels

Abstract

Communication enabled circuit breakers and circuit breaker panels are described. Methods associated with such communication enabled circuit breakers and circuit breaker panels are also described. A circuit breaker panel may include a circuit breaker controller and one or more communication enabled circuit breakers. Two-way wireless communication is possible between the circuit breaker controller and the one or more communication enabled circuit breakers.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5223795	12/1992	Blades	N/A	N/A
5436604	12/1994	Mrenna et al.	N/A	N/A
5467286	12/1994	Pyle et al.	N/A	N/A
5486755	12/1995	Horan et al.	N/A	N/A
5629869	12/1996	Johnson et al.	N/A	N/A
5805813	12/1997	Schweitzer, III	N/A	N/A
6034586	12/1999	Runyan	335/8	H01H 71/1045
6055144	12/1999	Reid	N/A	N/A
6195243	12/2000	Spencer et al.	N/A	N/A

6246928	12/2000	Louis et al.	N/A	N/A
6292717	12/2000	Alexander et al.	N/A	N/A
6295190	12/2000	Rinaldi et al.	N/A	N/A
6473794	12/2001	Guheen et al.	N/A	N/A
6628496	12/2002	Montjean	N/A	N/A
7493221	12/2008	Caggiano et al.	N/A	N/A
7546214	12/2008	Rivers, Jr. et al.	N/A	N/A
7558038	12/2008	Wiese et al.	N/A	N/A
7719257	12/2009	Robarge et al.	N/A	N/A
8463453	12/2012	Parsons, Jr.	N/A	N/A
8681463	12/2013	Franks et al.	N/A	N/A
8719123	12/2013	Rohrbaugh	N/A	N/A
9432214	12/2015	Lal et al.	N/A	N/A
9438026	12/2015	Franks et al.	N/A	N/A
9715796	12/2016	Reid	N/A	N/A
9853458	12/2016	Bell	N/A	N/A
10063592	12/2017	McClintock	N/A	N/A
2002/0075616	12/2001	Montjean	361/79	H02H 1/0061
2003/0086228	12/2002	Papallo, Jr. et al.	N/A	N/A
2003/0167373	12/2002	Winters et al.	N/A	N/A
2005/0116814	12/2004	Rodgers et al.	N/A	N/A
2005/0168891	12/2004	Nilman-Johansson	361/23	H01H 11/0062
2005/0225909	12/2004	Yoshizaki	N/A	N/A
2006/0050870	12/2005	Kimmel et al.	N/A	N/A
2006/0087783	12/2005	Holley	361/81	H02H 3/042
2006/0238932	12/2005	Westbrock et al.	N/A	N/A
2007/0143043	12/2006	Wafer et al.	N/A	N/A
2007/0188954	12/2006	Wiese	361/62	H02H 3/006
2007/0194942	12/2006	Darr	N/A	N/A
2008/0079437	12/2007	Robarge et al.	N/A	N/A
2008/0142486	12/2007	Vicente et al.	N/A	N/A
2008/0209429	12/2007	Van Riel	N/A	N/A
2008/0231485	12/2007	Newlin et al.	N/A	N/A
2008/0255782	12/2007	Bilac et al.	N/A	N/A
2009/0206059	12/2008	Kiko	N/A	N/A
2011/0150482	12/2010	Furusawa et al.	N/A	N/A
2011/0279933	12/2010	Campolo et al.	N/A	N/A
2012/0098347	12/2011	Beierschmitt et al.	N/A	N/A
2012/0123762	12/2011	Studer, II	N/A	N/A
2012/0140431	12/2011	Faxvog et al.	N/A	N/A
2012/0262093	12/2011	Recker et al.	N/A	N/A
2013/0029596	12/2012	Preston et al.	N/A	N/A
2013/0329331	12/2012	Erger	N/A	N/A
2014/0134951	12/2013	Paulson	N/A	N/A
2014/0146431	12/2013	Franks et al.	N/A	N/A
2014/0193294	12/2013	Kain et al.	N/A	N/A
2014/0197856	12/2013	Ostrovsky et al.	N/A	N/A

2014/0211345	12/2013	Thompson	N/A	N/A
2014/0300486	12/2013	Hummel et al.	N/A	N/A
2014/0340222	12/2013	Thornton et al.	N/A	N/A
2015/0162157	12/2014	Luebke et al.	N/A	N/A
2015/0168487	12/2014	Parker	N/A	N/A
2015/0188975	12/2014	Hansen et al.	N/A	N/A
2015/0207301	12/2014	Franks et al.	N/A	N/A
2015/0296599	12/2014	Recker et al.	N/A	N/A
2015/0338472	12/2014	Nuqui	N/A	N/A
2016/0099749	12/2015	Bennett et al.	N/A	N/A
2016/0100310	12/2015	Lee	N/A	N/A
2016/0163186	12/2015	Davidson	N/A	N/A
2016/0181036	12/2015	Langdon	N/A	N/A
2016/0225562	12/2015	Franks et al.	N/A	N/A
2016/0231375	12/2015	Roemer et al.	N/A	N/A
2016/0282828	12/2015	Jauquet et al.	N/A	N/A
2017/0005462	12/2016	Williams	N/A	H01H 71/74
2017/0064798	12/2016	Economy et al.	N/A	N/A
2017/0163023	12/2016	Niehoff	N/A	N/A
2018/0129801	12/2017	Cambou	N/A	N/A
2018/0278724	12/2017	Erdelyi	N/A	N/A
2020/0264234	12/2019	Miller et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
3002648	12/2015	EP	N/A
WO-2011091429	12/2010	WO	H02H 1/0015
WO-2014018434	12/2013	WO	H01H 71/123

OTHER PUBLICATIONS

Allen-Bradley, Rockwell Software Rockwell Automation, “Supplementary Protectors/Miniature circuit breakers” Catalog Nos. 1492-SP Series C, Apr. 2011, p. 31. cited by applicant
Elmark, “Technical Specification—Minature circuit breakers (MCB) C60DC Series”, pp. 3 and 4. cited by applicant
International Search Report and Written Opinion for Application No. PCT/US2018/019852 mailed on May 15, 2018, 13 pages. cited by applicant
International Search Report and Written Opinion for Application No. PCT/US2018/019856, mailed on May 11, 2018, 14 pages. cited by applicant
International Search Report and Written Opinion for Application No. PCT/US2018/019857, mailed on Jun. 27, 2018, 14 pages. cited by applicant
International Search Report and Written Opinion for Application No. PCT/US2018/019859, mailed on May 16, 2018, 11 pages. cited by applicant
Techterms, “Overwrite”, Techterms.com/definition/overwrite, Feb. 29, 2012. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of pending U.S. patent application Ser. No. 17/696,000, filed Mar. 16, 2022, entitled “Communication Enabled Circuit Breakers and Circuit Breaker Panels,”, which is a continuation application of U.S. patent application Ser. No. 16/485,491, filed Aug. 13, 2019, now U.S. Pat. No. 11,297,506, entitled “Communication Enabled Circuit Breakers and Circuit Breaker Panels,” which application is a United States National Phase filing of International Application No. PCT/US2018/019857, filed Feb. 27, 2018, which claims the benefit of U.S. Provisional Application Ser. No. 62/465,046, filed Feb. 28, 2017, entitled “Wireless Communication Enabled Circuit Breakers and Circuit Breaker Panels;” U.S. Provisional Application Ser. No. 62/500,051, filed May 2, 2017, entitled “Wireless Communication Enabled Circuit Breakers and Circuit Breaker Panels;” U.S. Provisional Application Ser. No. 62/612,654, filed Jan. 1, 2018, entitled “Secure Communication for Commissioning and Decommissioning Circuit Breakers and Panel System;” U.S. Provisional Application Ser. No. 62/612,656, filed Jan. 1, 2018, entitled “Communication Enabled Circuit Breakers;” and U.S. Provisional Application Ser. No. 62/612,657, filed Jan. 1, 2018, entitled “Communication Enabled Circuit Breakers;” which applications are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE
(1) The present invention relates generally to circuit breakers. More particularly, the present invention relates to communication enabled circuit breakers and circuit breaker panels that house circuit breakers.
BACKGROUND OF THE DISCLOSURE
(2) Circuit breakers provide protection in electrical systems by disconnecting a load from a power supply based on certain fault conditions, e.g., ground fault, arc fault, overcurrent. In general, circuit breakers monitor characteristics of the electrical power supplied to branch circuits. The circuit breakers function to interrupt, open, ‘trip’ or ‘break’ the connection between the power supply and a branch circuit when fault conditions (e.g., arc faults, ground faults, and unsafe overcurrent levels) are detected on the supplied branch, e.g., automatically open a switch to disconnect the branch from the power supply when such fault conditions are detected.
(3) Existing circuit breaker panels and circuit breakers housed by such panels may provide limited information to electricians and consumers about the nature of the fault conditions observed by the circuit breakers. For example, electricians and consumers may be able to determine that a circuit breaker has tripped by visual inspection of the circuit breaker or if power is lost on one or more loads. The visual inspection of the circuit breaker generally requires observing an operating switch associated with the circuit breaker. The operating switch of the circuit breaker is provided to allow for manually opening and closing contacts of the circuit breaker. The operating switch is also typically used to reset the circuit breaker after the circuit breaker has tripped due to a detected fault condition.
(4) It is to be appreciated, that circuit breakers are typically installed in circuit breaker panels, which are themselves typically located in dedicated electrical rooms, basements, garages, outdoor spaces, etc. Additionally, circuit breaker panels often include a door or cover that limits access to the circuit breakers housed therein. Therefore, locating, inspecting and/or resetting deployed circuit breakers may be difficult. Furthermore, because circuit breakers generally require visual inspection to determine if a fault condition has occurred, property owners and/or residents may not immediately recognize when an electrical fault condition has caused a circuit breaker to trip.

Failure to immediately recognize when an electrical fault condition has caused a circuit breaker to trip may cause damage to property and/or personal effects due to a loss of electricity to one or more loads.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates a communication enabled circuit breaker and panel system in accordance with an exemplary embodiment;

(2) FIG. **2** illustrates a communication enabled circuit breaker in accordance with an exemplary embodiment;

(3) FIG. **3** illustrates a circuit breaker controller in accordance with an exemplary embodiment;

(4) FIG. **4** illustrates a logic flow depicting communication between a circuit breaker controller and a communication enabled circuit breaker;

(5) FIG. **5** illustrates a logic flow associated with a circuit breaker controller;

(6) FIG. **6** illustrates a logic flow associated with a communication enabled circuit breaker; and

(7) FIG. **7** illustrates an embodiment of a storage medium.

DETAILED DESCRIPTION

(8) Communication enabled circuit breakers and circuit breaker panels are provided. Methods associated with such communication enabled circuit breakers and circuit breaker panels are also provided.

(9) Embodiments discussed herein, can provide a number of advantages over conventional devices. For example, the electrical components of the communication enabled circuit breaker is preferably advantageously supplied power from the line side phase and neutral electrical connections/terminations of the communication enabled circuit breaker, rather than the load side phase and neutral electrical connections/terminations typically used in conventional circuit breakers. Therefore, unlike conventional circuit breakers, a load side trip incident will not terminate power supplied to the electrical components, which includes the communication components of the circuit breaker. This is because some embodiments detailed herein provide communication components of communication enabled circuit breakers arranged to receive power from the line side phase and neutral. In the event of a trip incident, power is still available on the line side phase and neutral. Therefore, communication functionality is available even in the event of a trip incident.

(10) In general, the functionality of conventional circuit breakers is fixed at the time of manufacture. Specifically, circuit breakers are not upgradable or reprogrammable in the field. Therefore, in the event that a circuit breaker is to be upgraded due to a change in load conditions, or the like, physical access to the circuit breaker may be required. Such physical access to the circuit breaker may require dispatching an electrician to the site of the circuit breaker in order to allow for removing the circuit breaker and replacing it with a model which includes the desired functionality. Therefore, upgrading or reprogramming of circuit breakers may be too time-consuming or costly to implement.

(11) The embodiments detailed herein provide an advantage in that conditions in which circuit breakers interrupt faults can be updated and/or “customized” after manufacture, and even after initial deployment or installation. Accordingly, circuit breakers can be customized in the field and/or after installation to interrupt faults based on particular loads to which they are coupled and/or based on historical characteristics of the load. Thus, potentially reducing the number of unintentional fault interrupts that occur.

(12) It is noted, that the present disclosure often uses examples of communication enabled circuit breakers and panels, which may be wirelessly coupled. It is to be appreciated that the examples

given herein can be implemented using wired communication technologies (e.g., Ethernet, RS232, USB, or the like) instead of wireless communication technologies. As such, the use of the term “wireless” when referring to the communication technologies that may be implemented by the breakers and/or panels is not intended to be limiting to breakers and panels which only communicate wirelessly. Furthermore, system components can be referred to as “wireless” without implying that the elements recited thereto are devoid of wires or physical conductors/conductive paths.

(13) FIG. **1** illustrates a communication enabled circuit breaker and panel system **100** in accordance with an exemplary embodiment. The communication enabled circuit breaker and panel system **100** includes a circuit breaker panel **102**. The circuit breaker panel **102** may include any number of communication enabled circuit breakers **104**-*n*, where n is a positive integer. For example, system **100** is depicted including communication enabled circuit breakers **104-1**, **104-2**, **104-3**, **104-4**, **104-5**, **104-6**, **104-7**, **104-8**, **104-9** and **104-10**. It is noted, system **100** is depicted with communication enabled circuit breaker **104-1** to **104-10** for purposes of clarity and not limitation. For example, system **100** can include panel **102** having any number (e.g., 1, 2, 3, 4, or more) of communication enabled circuit breakers **104**-*n*. Additionally, panel **102** may include both communication enabled circuit breakers (e.g., **104-1** to **104-10**) as well as conventional circuit breakers (not shown).

(14) Additionally, although each of the communication enabled circuit breakers **104-1** to **104-10** are labeled as breaker **104**, it is to be understood that communication enabled circuit breakers **104-1** to **104-10** are not necessarily identical. For example, communication enabled circuit breaker **104-1** may be a ground fault circuit interrupter (GFCI) device; communication enabled circuit breaker **104-2** may be an arc fault circuit interrupter (AFCI) device; communication enabled circuit breaker **104-3** may be a conventional overcurrent circuit breaker, an overcurrent hydraulic-magnetic circuit breaker, an overcurrent thermal magnetic circuit breaker, or the like; communication enabled circuit breaker **104-4** may include both GFCI and AFCI functionalities. Furthermore, each of the communication enabled circuit breakers **104-1** to **104-10** may be rated for a predefined trip amperage or overcurrent state, and not necessarily the same predefined trip amperage or overcurrent state.

(15) Furthermore, communication enabled circuit breakers **104-1** to **104-10** may be shaped and sized differently. For example, communication enabled circuit breaker **104-1** may be a double pole circuit breaker having a 2 inch width; communication enabled circuit breaker **104-2** may be a single circuit breaker having a 1 inch width; communication enabled circuit breaker **104-2** may be a circuit breaker having a ¾ inch width; communication enabled circuit breaker **104-2** may be a circuit breaker having a 1½ inch width; etc. The width of the communication enabled circuit breakers **104-1** to **104-10** refers to the shorter side of the generally rectangular visible face of the wireless circuit breakers **104-1** to **104-10** once it is installed in the circuit breaker panel **102**.

(16) Each of the communication enabled circuit breakers **104-1** to **104-10** may include communication components (refer to FIG. **2**), which in some examples can be wireless. Such communication components associated with each of the communication enabled circuit breakers **104-1** to **104-10** may enable the communication enabled circuit breakers **104-1** to **104-10** to communicate (e.g., send and/or receive information elements including data, indications of operating conditions, instructions, updated fault interruption instructions, or the like) using any of a variety of communication standards. For example, in the case of wireless communication, the communication enabled circuit breakers **104-1** to **104-10** can include wireless communication components arranged to communicate via near field or personal area network communications protocols, e.g., Bluetooth® Low Energy (BLE) technology, thus enabling the communication enabled circuit breakers **104-1** to **104-10** to communicate using BLE communication schemes. In some implementations, the use of BLE is advantageous verses other wireless communication standards, such as standard Bluetooth®, since BLE requires less power, thereby minimizing the space required within the communication enabled circuit breakers **104** to house the circuitry. In the

case of wired communication, the communication enabled circuit breakers **104-1** to **104-10** can include wired communication components arranged to communicate via a wired communication protocol, e.g., USB or MTP, thus enabling the wired circuit breakers to communicate using a wired communication scheme.

(17) The circuit breaker panel **102** further houses a circuit breaker controller **106**. The circuit breaker controller **106** may include communication components (refer to FIG. **3**). In an alternative embodiment, the circuit breaker controller **106** is coupled to the circuit breaker panel **102** in an external arrangement. For example, the controller **106** could be housed in a different panel than panel **102** or disposed external to panel **102**. The communication components associated with the circuit breaker controller **106** may enable the controller **106** to communicate (e.g., send and/or receive information elements including data, indications of operating conditions, instructions, updated fault interruption instructions, or the like) using any of a variety of communication standards.

(18) For example, the wireless circuit breaker controller **106** can include wireless communication components arranged to communicate via BLE technology, thus enabling the wireless circuit breaker controller **106** to communicate using BLE communication schemes. Accordingly, the wireless circuit breaker controller **106** can communicate with the wireless circuit breakers **104** wirelessly, for example, using BLE communication schemes.

(19) In general, the communication enabled circuit breakers **104-1** to **104-10** and the circuit breaker controller **106** (and particularly, the communication components of these devices) can be arranged to communicate using a variety of communication technologies, which may be wireless or wired in nature. For example, the circuit breaker controller **106** can be arranged to wirelessly communicate via ZigBee®, Z-Wave, Bluetooth®, Bluetooth® Low Energy (BLE), 6LowPan, Thread, Cellular, Sigfox®, NFC, Neul®, LoRaWAN™, or the like. In some implementations, the communication enabled circuit breakers **104** and circuit breaker controller **106** may communicate via wired (as opposed to wireless) technologies. For example, the communication enabled circuit breakers **104** may be communicatively coupled via a wired link to the circuit break controller **106**.

(20) The circuit breaker controller **106** may be configured to communicate via multiple communication components. For example, circuit breaker controller **106** may be configured to communicate with communication enabled circuit breakers **104** via BLE as described above. Additionally, the circuit breaker controller **106** can be configured to communicate (e.g., send and/or receive information elements including data, indications of operating conditions, instructions, updated fault interruption instructions, or the like) via a second wireless communication scheme or via a wired communication scheme. For example, the circuit breaker controller **106** could include wireless communication components arranged to communicate via Wi-Fi technology, thus enabling the circuit breaker controller **106** to communicate using Wi-Fi communication schemes. Accordingly, the circuit breaker controller **106** can communicate with devices external to the circuit breaker panel **102** via wireless channel **108**, for example, using Wi-Fi communication schemes. In general, however, the circuit breaker controller **106** may be enabled to communicate with devices external to the circuit breaker panel **102** using any suitable type of communication technology, either wireless or wired (e.g., BLE, 4G, LTE, Wi-Fi, USB, RS232, MTP, etc.).

(21) Components from the circuit breaker panel **102** may communicate (e.g., wirelessly or wired) with one or more remote entities **120**. For example, the communication enabled circuit breakers **104** and/or the circuit break controller **106** of the circuit breaker panel **102** may communicate wirelessly with a mobile device **110** (e.g., tablet computer, mobile phone, etc.), a computing device **112** (desktop computer, server, etc.) and/or the Internet cloud **114**. For example, the communication enabled circuit breakers **104-1** to **104-10** can communicate with the circuit breaker controller **106**, which can itself, communication with any one of remote entities **120**. It is noted, remote entities **120** are depicted including mobile device **110**, computing device **112**, and Internet **114**. However, remote entities **120** could include just a single device or entity remote to circuit breaker panel **120**.

The term remote entities **120** is used herein to refer to one or more devices remote to the panel **120**, such as, for example, mobile device **110**, computing device **112**, and Internet **114**. Furthermore, although the term remote entity **120** is sometimes used herein in the plural, it is not intended to imply or denote multiple devices or multiple entities remote to panel **102** but could simply refer to a single entity remote to the system (e.g., just the Internet **114**, just the mobile device **110**, or the like).

(22) More particularly, the circuit breakers **104** can wirelessly communicate with the circuit breaker controller **106**, which can itself, wirelessly communicate with devices remote to the circuit breaker panel. In some examples, the wireless circuit breakers **104** can directly couple to devices remote to the circuit breaker panel **102**. For example, the mobile device **110** can communicate directly (e.g., via BLE) with one of the wireless circuit breakers **104**. In addition, the circuit breaker panel **102** may include wireline connectivity functionality, such as an Ethernet port, to enable wireline communication with one or more remote entities.

(23) In some examples, the communication enabled circuit breakers **104-1** to **104-10** can directly couple to remote entities **120**. For example, the mobile device **110** can communicate directly (e.g., via BLE) with at least one of the communication enabled circuit breakers **104-1** to **104-10**. In addition, the circuit breaker panel **102** (e.g., via the circuit breaker controller **106**) may include wireline connectivity functionality, such as an Ethernet port, to enable wireline communication with one or more remote entities. In some implementations, the communication enabled circuit breakers **104-1** to **104-10** may establish a mesh network. For example, communication enabled circuit breaker **104-1** may share a wireless connection with a remote entity **120** with communication enabled circuit breaker **104-2**. Furthermore, in such a mesh network topology, communication enabled circuit breaker **104-2** may share the wireless connection to the remote entity **120** with communication enabled circuit breaker **104-3** and communication enabled circuit breaker **104-4**. Therefore, using the mesh network topology, the wireless connection to the remote entity **120** may be shared between the communication enabled circuit breakers **104-1** to **104-10**. The mesh network may be implemented in accordance with wireless communication schemes, or standards, such as, BLE standards, Wi-Fi standards, or the like.

(24) The present disclosure provides that the communication enabled circuit breakers **104** and the circuit breaker controller **106** may communicate, or exchange signals including indications of data, operating conditions, fault detection events, fault interruption instructions, or the like. For example, the circuit breaker controller **106** may be configured to transmit updated software (e.g., operating software, firmware, fault interrupter instructions, etc.) to one or more of the communication enabled circuit breakers **104**. For example, the circuit breaker controller **106** may provide, e.g., updated firmware to one or more of the communication enabled circuit breakers **104**. Furthermore, the circuit breaker controller **106** may provide updated fault interrupter instructions to one or more of the communication enabled circuit breakers **104**. The updated fault interrupter instructions may replace fault interrupter instructions stored in one or more of the communication enabled circuit breakers **104**. The updated fault interrupter instructions may be received at the circuit breaker controller **106** from one or more remote entities **120** (e.g., mobile device **110**, computing device **112** and/or the Internet cloud **114**). The circuit breaker controller **106** may communicate updated fault interrupter instructions to one or more of the communication enabled circuit breakers **104** (e.g., using wireless or wired communication technologies). Alternatively, one or more remote entities **120** (e.g., mobile device **110**, computing device **112** and/or the Internet cloud **114**) may directly provide updated fault interrupter instructions to one or more of the communication enabled circuit breakers **104**. To that end, the one or more remote entities **120** may communicate fault interrupter instructions to one or more of the communication enabled circuit breakers **104** (e.g., using BLE, or the like). The process of providing updated firmware and/or updated fault interrupter instructions may also be provided to other interrupter devices, such as AFCI/GFCI receptacles.

(25) In general, fault interrupter instructions stored in the communication enabled circuit breaker

104 and/or updated fault interrupter instructions may include parameters of operating conditions intended to trigger a trip event. Said differently, the fault interrupter instructions and/or updated fault interrupter instructions can be configured to control fault condition detection algorithms and/or fault interruption characteristics of the communication enabled circuit breakers **104**. For example, the fault interrupter instructions stored in the communication enabled circuit breaker **104** and/or updated fault interrupter instructions may define an overcurrent trip value in amps and an overcurrent trip response time in seconds. In addition, the fault interrupter instructions stored in the communication enabled circuit breaker **104** and/or updated fault interrupter instructions may define parameters associated with the communication enabled circuit breaker **104**.

(26) Such parameters may include current rating, voltage rating, time current curve characteristics (e.g., the relationship between the sensed overcurrent and the time required under which to trip the communication enabled circuit breaker **104**), status, trip alarm, remote trip, single phase or three phase, and the like. In a particular implementation, the fault interrupter instructions stored in the communication enabled circuit breaker **104** and/or updated fault interrupter instructions may also include power metering instructions. The power metering instructions may enable the communication enabled circuit breaker **104** to measure energy, line and/or phase voltages, line frequency, line and/or phase current, and/or power consumption.

(27) In one embodiment, the circuit breaker controller **106** includes a power supply that is coupled to a line side phase connection. Likewise, each of the communication enabled circuit breakers **104** includes a power supply that is coupled to a line side phase connection of the communication enabled circuit breaker (e.g., before the set of interruptible contacts of the communication enabled circuit breaker **104**). The power supply may be a AC to DC converter, an AC to AC converter, or the like. In general, the power supply is provided to condition and/or convert a voltage of the line side phase and neutral electrical connections to one or more voltages that are supplied to components of the circuit breaker controller **106** and/or each of the communication enabled circuit breakers **104**. Furthermore, the power supply may include one or more fuses to protect components of the circuit breaker controller **106** and/or each of the communication enabled circuit breakers **104**. Advantageously, the circuit breaker controller **106** and/or each of the communication enabled circuit breakers **104** are supplied with power even in the event of a load side incident that causes one or more of the communication enabled circuit breakers **104** to trip.

(28) In one embodiment, one or more of the communication enabled circuit breakers **104** are configured to automatically wirelessly transmit fault related information upon occurrence of a trip incident. In particular, one or more of the communication enabled circuit breakers **104** wirelessly transmits such fault related information to the circuit breaker controller **106**. In one embodiment, the fault related information is transmitted using BLE. The fault related information may include, for example, recorded parameters that may have caused the trip incident, parameters used to determine that tripping the communication enabled circuit breaker **104** was necessary, a unique identifier of the communication enabled circuit breaker **104** that tripped (e.g., a unique wireless circuit breaker serial number, or the like), a time and date of the trip incident, a load signature that caused the trip incident, a zone or area that is without power as a result of the trip incident, etc. The circuit breaker controller **106** may disseminate the fault related information to one or more devices external of the circuit breaker panel **102**. For example, the circuit breaker controller **106** may wirelessly transmit the fault related information to one of the remote entities **120**. Therefore, a user or users of the remote entities **120** may be made immediately aware of the trip incident by way of at least the transmitted fault related information.

(29) In another embodiment, one or more of the communication enabled circuit breakers **104** are configured to transmit status related information upon request by the circuit breaker controller **106**. The request by the circuit breaker controller **106** may be communicated to one or more of the wireless circuit breakers **104** using any of the techniques discussed herein (e.g., wireless or wired). The status related information may include self-test related information provided by one or more of

the communication enabled circuit breakers **104**. In one implementation, the self-test related information may include operational status of components associated with one or more of the communication enabled circuit breakers **104**.

(30) As discussed above, one or more remote entities **120** (e.g., mobile device **110**, computing device **112** and/or the Internet cloud **114**) may directly communicate with one or more of the communication enabled circuit breakers **104**. For example, the mobile device **110** may provide updated fault interrupter instructions to one or more of the communication enabled circuit breakers **104**. In such an embodiment, the one or more remote entities **120** may communicate (e.g., via BLE) updated fault interrupter instructions directly to one or more of the communication enabled circuit breakers **104** without communicating through the circuit breaker controller **106**. Direct communication between a remote entity **120** and the communication enabled circuit breakers **104** enables an electrician (or other suitable user) to interface in real-time with the communication enabled circuit breakers **104** and conduct a number of operations, all from within the same application on the remote entity. For example, the electrician may put a communication enabled circuit breaker **104** in a data acquisition/logging only mode, where when the communication enabled circuit breaker **104** is in the data acquisition only mode, the communication enabled circuit breaker **104** will not trip upon the occurrence of a fault but, rather, would pass the data in real-time to the remote entity **120** for analysis by either the remote entity **120**, the electrician, or other suitable party.

(31) For example, in the data acquisition only mode, the communication enabled circuit breaker **104** may be configured to not trip upon the occurrence of an arc fault, a ground fault, an overcurrent fault, or a combination of these. In other words, in the data acquisition only mode, the communication enabled circuit breaker **104** may be configured to trip upon detection of an overcurrent but not trip upon detection of an arc fault. In addition to the electrician, remote users may also communicate with the communication enabled circuit breaker **104** and/or the electrician via the same application on different devices. Additionally, the electrician can then iteratively make adjustments to the fault interrupter instructions and transmit updated fault interrupter instructions to the communication enabled circuit breaker **104** and analyze the resulting data from the communication enabled circuit breaker **104**. In this manner, the fault interrupter instructions can be refined/tailored to avoid unintentional fault interrupts (e.g., nuisance tripping) of the communication enabled circuit breaker **104**. This refinement of the fault interrupter instructions may be done for any particular communication enabled circuit breaker **104** as required, a plurality of communication enabled circuit breakers **104**, or every communication enabled circuit breaker **104** in the circuit breaker panel **102**. Likewise, the refinement of the fault interrupter instructions may be implemented across multiple installations or be implemented globally to all suitable breakers and in many or all installations. While the communication enabled circuit breaker **104** is in the data acquisition only mode, the communication enabled circuit breaker **104** may indicate, by any number of methods, when it would otherwise trip. Examples of such indication include flashing of LED(s) or transmitting the indication to the remote entity **120**.

(32) As described above, the present disclosure can be implemented to provide communication enabled circuit breakers **104**, circuit breaker controller **106**, and remote entities **120**, arranged to communicate via either wired or wireless communication protocols and technologies. However, for clarity of presentation, the following examples depict and describe communication enabled circuit breakers **104** and a circuit breaker controller **106** arranged to communicate via wireless communication protocols. As such, many of the communication enabled circuit breakers **104** described in the following examples are referred to as “wireless circuit breakers” **104** or “communicating circuit breaker” **104**. Likewise, the circuit breaker controller **106** may be referred to as a “wireless circuit breaker controller” **106**. This is not intended to be limiting and the example breakers, controller, remote entities, techniques, and systems depicted and described below can be implemented with wired communication technologies without departing from scope of the

disclosure. Additionally, the wireless circuit breakers **104** and the wireless circuit breaker controller **106** are described herein to communicate via BLE for purposes of convenience and clarity of presentation. This is also not intended to be limiting.

(33) FIG. **2** illustrates the wireless circuit breaker **200** in accordance with an exemplary embodiment. In some examples, the wireless circuit breaker **200** can be implemented as any one of the communication enabled circuit breakers **104-1** to **104-1** of the system **100** of FIG. **1**. Generally, the wireless circuit breaker **200** may be used in a wide range of commercial, residential, and industrial circuit breaker panels. The wireless circuit breaker **200** may be configured to operate in conjunction with different electrical power distribution systems, including single-phase, split-phase, 3-phase delta, and 3-phase star. These systems may operate at any suitable voltage such as 120/240 (120V phase-neutral, 240 phase-to-phase), 120/208, 265/460, 277/480.

(34) The wireless circuit breaker **200** includes multiple connections or “terminals.” Specifically, wireless circuit breaker **200** includes a line side phase connection **202**, a line side neutral connection **203**, a load side phase connection **204**, and a load side neutral connection **205**. The line side phase connection **202** and line side neutral connection **203** are coupled to a power source. The load side power phase connection **204** and load side neutral connection **205** are coupled to a load. Thus, current can enter the wireless circuit breaker **200** via the line side phase connection **202**, exit the wireless circuit breaker **200** via the load side phase connection **204**, return to the wireless circuit breaker **200** via load side neutral connection **205**, and travel back to the power source via line side neutral connection **203**. The line side phase connection **202** and neutral connection **203** may be coupled to a power source (e.g., an electrical grid). The load side phase connection **204** and the load side neutral connection **205** may be coupled to branch circuit that may feed a load (e.g., HVAC system, refrigerator, TV, etc.).

(35) The wireless circuit breaker **200** may include a power supply **206**. The power supply **206** receives an input power from the line side phase connection **202** and line side neutral connection **203**. The power supply **206** converts, in one implementation, an AC voltage to a regulated DC voltage for use by some or all of the electrical components associated with the wireless circuit breaker **200**. To that end, the voltage provided by the power supply **206** is uninterrupted even when the wireless circuit breaker **200** is caused to trip as a result of a trip incident. In some examples, the power supply **206** includes circuitry to condition the current and/or voltage supplied to the electrical components of the wireless circuit breaker **200**. In some examples, power supply **206** includes a fuse, which can in some embodiments be replaceable, to protect the power supply **206** and wireless circuit breaker **200** from overcurrent conditions. In some examples, power supply **206** itself includes a circuit breaker to protect the power supply **206** and wireless circuit breaker **200** from overcurrent conditions. With some examples, power supply **207** can be arranged to compensate for various electrical conditions that may be present on the input line for the panel system **102**. For example, power supply **207** could be arranged to compensate for under-voltage conditions, filter interference, or the like.

(36) A memory **208** is disposed in the wireless circuit breaker **200**. The memory **208** is configured to store fault interrupter instructions **210**. The memory **208** may comprise an article of manufacture. In some examples, the memory **208** may include any non-transitory computer readable medium or machine readable medium, such as an optical, magnetic or semiconductor storage. The memory **208** may store various types of computer executable instructions, such as the fault interrupter instructions **210**. The memory **208** may be coupled to a processor **212**. Processor **212** could be any of a variety of processors, such as, for example, a central processing unit, a microprocessor, a field programmable gate array, an application specific integrated circuit, or the like. Processor **212** can be arranged to execute fault interrupter instructions **210** to aid in performing one or more techniques described herein (e.g., cause the wireless circuit breaker **200** to trip, cause the wireless circuit breaker **200** to transmit information pertaining to a trip incident, etc.).

(37) The wireless circuit breaker **200** includes a fault interrupter **214** or a “circuit interrupter” **214**. In one implementation, the fault interrupter **214** is operable to interrupt faults (e.g., decouple the load side phase connection **204** from the line side phase connection **202**) based in part on the fault interrupter instructions **210** stored in the memory **208**. As used herein, the term “fault” could include any of a variety of conditions with which it may be desirable for the wireless circuit breaker **200** to disconnect the line side connection from the load side connection. For example, “fault” may be a fault within the breaker, a fault on the load side, a fault on the line side, or the like. As another example, “fault” may be a ground fault, an arc fault, an overcurrent fault, or the like. Examples are not limited in these contexts. The fault interrupter **214** may comprise various hardware elements. In one example, the fault interrupter **214** includes at least a trip solenoid and/or an energy storage element to trip the trip solenoid and cause the line side connection **202** to decouple from load side connection **204**. In further examples, the fault interrupter **214** can include a reset solenoid and/or energy storage element to set the breaker **200** and cause the line side connection **202** to couple to the load side connection **204**.
(38) The fault interrupter instructions **210** may be executed (e.g., by fault interrupter **214**, by processor **212**, or the like) to cause the trip solenoid to break current flowing from the line side phase connection **202** to the load side phase connection **204** in specific conditions. For example, when the current exceeds a threshold defined by the fault interrupter instructions **210**. In another example, the fault interrupter **214** includes functionality, controllable by way of the fault interrupter instructions **210**, to sense characteristics of a line current, for example an amount of current, a frequency of the current, high-frequency current components, dynamic distribution of the frequency components over time and within a half cycle of a power line frequency, various profiles of power line characteristics, etc. As another example, the fault interrupter **214** includes functionality, controllable by way of the fault interrupter instructions **210**, to set the breaker **200**, such as, upon receipt of a control signal from a remote entity **120** where the control signal includes an indication to set the breaker.
(39) The fault interrupter **214** may be sensitive to radio frequency (RF) signals (i.e., wireless signals). Therefore, the fault interrupter **214** may be partially or completely surrounded by an RF shielding **216**. The RF shielding **216** may comprise any suitable material to attenuate wireless signals, for example, any ferrous material. In one implementation, the RF shielding **216** shields the fault interrupter **214** from wireless signals generated by the: wireless circuit breaker **200**, other wireless circuit breakers **200**, circuit breaker controller **106**, and/or entities external of the circuit breaker panel **102** (e.g., remote entities **120**, or the like).
(40) The wireless circuit breaker **200** includes wireless communication components **218**. The wireless communication components **218** enables the wireless circuit breaker **200** to communicate wirelessly using any suitable type of wireless communication technology as described herein. Therefore, the wireless communication components **218** may include at least a radio **226**, antenna **224**, and processor **222**. In general, the radio **226** can be any radio configured to communicate using a wireless transmission scheme, such as, for example, BLE. The antenna **224** can be coupled to radio **224** and configured to emit and receive RF signals. For example, the antenna **224** can emit RF signals received from the radio **226** coupled between the radio **226** and the antenna **224**. The antenna **224** could be any of a variety of antennas (or antenna arrays) having different shapes and/or configurations arranged to emit/receive radio waves on a particular frequency, range of frequencies, or the like. Furthermore, the antenna **224** could be internal to the housing **201** of the wireless circuit breaker **200** or external to the housing **201** of the wireless circuit breaker **200**. Processor **222** can be any of a variety of processors (e.g., application processor, baseband processors, etc.) arranged to perform at least transmission and reception of wireless signals associated with the wireless circuit breaker **200**.
(41) As described, the wireless communication components **218** receives power from the power supply **206**, which is coupled to the line side phase connection **202** and line side neutral connection

203. Therefore, the wireless communication components **218** enable the wireless circuit breaker **200** to communicate wirelessly even in the event that the fault interrupter **214** interrupts current flowing between the line side phase connection **202** and the load side phase connection **204**.

(42) An indicator may be implemented on the wireless circuit breaker **200**. The indicator may be any suitable type of indicator such as a visual or audible indicator including but not limited to, an LED, neon bulb, and/or piezoelectric buzzer. In the present embodiment, the indicator is a light emitting diode (LED) **220**. The LED **220** may be illuminated to a predefined color, illumination pattern, and/or illumination frequency, when the wireless circuit breaker **200** is in an update mode. The update mode indicates that the wireless circuit breaker **200** is ready to receive updated fault interrupter instructions for storage in the memory **208** from a remote entity **120**. In some implementations, when the wireless circuit breaker **200** is an update mode, the wireless circuit breaker **200** is open or tripped. In some implementations, when the wireless circuit breaker **200** is an update mode, the wireless circuit breaker **200** is unable to provide tripping functionality.

(43) The wireless circuit breaker **104** may comprise a housing **201**. The housing **201** may be a miniature circuit breaker (MCB) housing. In some implementations, the MCB housing has a width of 1 inch. It is noted, that the dimensions of the breakers are given for example only. Breaker widths could be any width, e.g., ½ inch, ¾ inch, 1 inch, 1½ inches, 2 inches, or the like.

(44) As noted above, with some implementations, wireless circuit breaker **200** can include a “diagnostic mode” or a data acquisition only mode. In such a mode, the wireless circuit breaker **200** may be configured to not trip upon occurrence of a selected faults. For example, the wireless circuit breaker **200** can be a combination AFCI/GFCI breaker. However, upon initialization of the data acquisition only mode, the breaker may be arranged to trip upon occurrence of a GFCI fault but not an AFCI fault. The wireless circuit breaker **200** can be arranged to log data (e.g., save indications of operating conditions, or the like) to memory **208**. Such logged data may be transmitted to a remote entity (e.g., via circuit breaker controller, or the like) and used by a manufacturer or technician to diagnose false AFCI tripping and/or nuisance tripping issues and to generate and/or develop new firmware (which can include fault interruption instructions) for wireless circuit breaker **200**.

(45) With some examples, wireless circuit breaker **200** could be deployed as a “diagnostic breaker,” which may include a handle (like the reset switch of conventional circuit breakers) to place the wireless circuit breaker **200** in the diagnostic mode. Furthermore, the wireless circuit breaker **200** may include multiple indications (e.g., LEDs **220**, or the like). For example, wireless circuit breaker **200** could include an LED **220** arranged to illuminate when the breaker is in a normal mode (e.g., arranged to trip on AFCI, GFCI, or both AFCI and GFCI). As a specific example, the wireless circuit breaker **200** could include an LED **220** arranged to illuminate when the breaker is configured to monitor and trip for GFCI and another LED **220** arranged to illuminate when the breaker is configured to monitor and trip for AFCI. As another example, the wireless circuit breaker **200** could include an LED **220** to indicate when the breaker is in a diagnostic mode.

(46) FIG. **3** illustrates the wireless circuit breaker controller **300** in accordance with an exemplary embodiment. In some examples, the wireless circuit breaker controller **300** can be implemented as the circuit breaker controller **106** of the system **100** of FIG. **1**. Generally, the wireless circuit breaker controller **300** may be used a wide range of commercial, residential, and industrial circuit breaker panels. In one embodiment, the wireless circuit breaker controller **300** is implemented in the circuit breaker panel **102**. In an alternative embodiment, the wireless circuit breaker controller **300** is coupled to the circuit breaker panel **102** in an external arrangement. For example, in an alternative implementation of the wireless circuit breaker controller **300**, the wireless circuit breaker controller **300** is part of a mobile device, such as a mobile phone, having hardware/software functionality to enable the mobile device to function as the described wireless circuit breaker controller **300**.

(47) A memory **302** is disposed in the wireless circuit breaker controller **300**. The memory **302** is

configured to store updated fault interrupter instructions **304**. The memory **302** may comprise an article of manufacture. In some examples, the memory **302** may include any non-transitory computer readable medium or machine readable medium, such as an optical, magnetic or semiconductor storage. The memory **302** may store various types of computable executable instructions, such as the updated fault interrupter instructions **304**.

(48) The memory **302** may be coupled to a processor **306**. Processor **306** could be any of a variety of processors, such as, for example, a central processing unit, a microprocessor, a field programmable gate array, an application specific integrated circuit, or the like. Processor **306** can be arranged to execute instructions stored in the memory **302** to aid in performing one or more techniques described herein (e.g., cause the updated fault interrupter instructions **304** to be sent to one or more of the wireless circuit breakers **200**, receive fault information including unique identifiers associated with wireless circuit breakers **200** and a time and date of a trip incident that caused fault interrupters **214** to interrupt the current flow between line side phase connection **202** and load side phase connection **204**, etc.).

(49) The wireless circuit breaker controller **300** may include a power supply **308**. The power supply **308** is to convert, in one implementation, an AC voltage to a regulated DC voltage for use by some or all of the electrical components associated with the wireless circuit breaker controller **300**. With some examples, power supply **308** can include multiple “hot” terminals and a neutral terminal. Thus, power supply **308** could receive power from either “hot” wire to provide redundancy. In the case of multi-phase systems, the power supply **308** could be arranged to couple to multiple phases to provide redundancy for the loss of one of phases.

(50) The wireless circuit breaker controller **300** includes wireless communication components **310**. The wireless communication components **310** enable the wireless circuit breaker controller **300** to communicate wirelessly using any suitable type of wireless communication technology (e.g., a short-range wireless/near field wireless technology, Bluetooth®, Wi-Fi, ZigBee®, etc.). Therefore, the wireless communication components **310** may include at least radio **318-1**, which may include radio and transmitting and receiving circuitry, antenna **316-1**, and processor **314-1**. In general, the radio **318-1** can be any radio configured to communicate using a wireless transmission scheme, such as, for example, BLE. The antenna **316-1** can be coupled to radio **318-1** and configured to emit and receive RF signals. For example, the antenna **316-1** can emit RF signals received from the radio **318-1** coupled between the radio **318-1** and the antenna **316-1**. The antenna **316-1** could be any of a variety of antennas (or antenna arrays) having different shapes and/or configurations arranged to emit/receive radio waves on a particular frequency, range of frequencies, or the like. Processor **314-1** can be any of a variety of processors (e.g., application processor, baseband processors, etc.) arranged to perform at least transmission and reception of wireless signals associated with the wireless circuit breaker controller **300**. Furthermore, the antenna **316-1** could be internal to the physical housing of the wireless circuit breaker controller **300** or external to the housing of the wireless circuit breaker controller **300**.

(51) As detailed, some embodiments provide wireless communication components **310** of wireless circuit breaker controller **300** can be operable communicate over a number of wireless frequencies or schemes. As such, processor **314-1**, radio **318-1** and antenna **316-1** could be arranged to communicate over multiple wireless communication technologies, such as, for example, BLE and Wi-Fi. In other examples, wireless communication components **310** can include multiple sets of processor, radio and antenna. For example, as depicted, components **310** further include radio **318-2**, antenna **316-2** and processor **314-2**. Thus, the first set of radio **318-1**, antenna **316-1** and processor **314-1** can be arranged to communicate using a first wireless communication scheme, such as, BLE while the second set of radio **318-2**, antenna **316-2** and processor **314-2** can be arranged to communicate using a second wireless communication scheme, such as, Wi-Fi.

(52) The wireless circuit breaker controller **300** may further include a wireline network interface **312**. The wireline network interface **312** enables the wireless circuit breaker controller **300** to be

coupled via a wireline connection to various devices. For example, in one implementation, the wireless circuit breaker controller **300** is a standalone device that may be wireline connected (e.g., via Ethernet) to a remote device (e.g., Internet cloud **114**) and wirelessly connected to breakers **104** within the circuit breaker panel **102**. In such an example, the controller **300** could optionally omit one of the wireless communication components (e.g., wireless communication components **310** arranged to communicate via Wi-Fi, or the like). As another example, the wireless circuit breaker controller **300** could be wireless coupled to wireless circuit breakers (e.g., wireless circuit breaker **200**, or the like) via wireless communication components **310** and coupled via a wired communication connection to other communication enabled circuit breakers (not shown) via wireline network interface **312**.
(53) FIGS. **4-6** illustrate logic flows, implementable by a communication enabled circuit breaker and panel system, such as, the system **100** of FIG. **1**. In general, these logic flows can be implemented by any communication enabled circuit breaker and panel system or component(s) of such a system, such as, the system **100**, communication enabled circuit breakers **104-*n***, circuit breaker controller **106**, remote entity **120**, communication enabled circuit breaker **200**, circuit breaker controller **400**, and/or the like. The following description of FIGS. **4-6** reference remote entity **120**, controller **300** and breaker **200** for purposes of convenience and clarity only. However, it is to be understood that the logic flows described could be implemented by different combinations of components of a communication enabled circuit breaker and panel system without departing from the spirit and scope of the claimed subject matter.
(54) Furthermore, it is noted that the operations for the depicted logic flows may occur sequentially and/or be made in parallel. Additionally, the operations for the depicted logic flows are not illustrated in a required order, unless dictated by the context or specifically states herein. That is, a different order other than that illustrated may be used. Some or all of the communications and operations associated with the logic flows may be embodied as one or more computer executable instruction. Such computer executable instructions may be stored in the storage medium, such as the memory **208** and/or the memory **302**. A computing device, such as the processor **212** and/or the processor **306**, may execute the stored computer executable instructions to implement the logic flows or cause the respective devices to implement the logic flows.
(55) FIG. **4** illustrates a logic flow **400** depicting communication between a wireless circuit breaker controller **300** and a wireless circuit breaker **200**. The logic flow **400** may provide updated fault interrupter instructions that may replace fault interrupter instructions stored in a memory. The left side of FIG. **4** shows communications transmitted and received by the wireless circuit breaker controller **300** while the right side of FIG. **4** shows communications transmitted and received by the wireless circuit breaker **200**.
(56) The logic flow **400** may begin with communication **402**. However, the logic flow **400** may begin with a different communication other than the communication **402**. At communication **402**, the wireless circuit breaker controller **300** sends an update control signal to the wireless circuit breaker **200**. The update control signal is to cause the wireless circuit breaker **200** to enter an update mode **404**. For example, the update control signal may cause the wireless circuit breaker **200** to open or trip. Alternatively, the update control signal may cause the wireless circuit breaker **200** to disable tripping functionality while leaving the wireless circuit breaker **200** in the “closed” position. Furthermore, the wireless circuit breaker **200** may illuminate and associated LED (e.g., the LED **220**) in the update mode. In one implementation, the LED will flash during the update mode. For example, the LED may flash red during the update mode and then transition to a solid green after the update mode is complete.
(57) At communication **406**, the wireless circuit breaker **200** communicates an update ready signal. The update ready signal indicates that the wireless circuit breaker **200** is ready to receive updated information, such as updated fault interrupter instructions. In some examples, the wireless circuit breaker **200** may only communicate an update ready signal if the wireless circuit breaker **200** is in

the open position (e.g., update mode properly entered at **404**, etc.). In one embodiment, the update ready signal may be communicated to include wireless circuit breaker **200** information that pertains to the wireless circuit breaker **200**. In one implementation, the wireless circuit breaker information includes at least a unique identifier of the wireless circuit breaker **200** and information indicating functionalities associated with the wireless circuit breaker **200**. For example, the information may identify that the wireless circuit breaker **200** is an AFCI, GFCI, or AFCI/GFCI, etc., wireless circuit breaker. The wireless circuit breaker information (e.g., the wireless circuit breaker type) may be used by the wireless circuit breaker controller to select appropriate updated fault interrupter instructions for communication to the wireless circuit breaker **200**.

(58) At communication **408**, the wireless circuit breaker controller **300** communicates updated fault interrupter instructions to the wireless circuit breaker **200**. At block **410**, the wireless circuit breaker **200** validates the updated fault interrupter instructions received from the wireless circuit breaker controller **300**. In one implementation, the validation process involves a checksum verification process in which a checksum associated with the updated fault interrupter instructions is compared against a checksum associated with fault interrupter instructions previously stored in the wireless circuit breaker **200**. If the validation fails, the circuit breaker may not use the updated fault interrupter instructions to replace the fault interrupter instructions previously stored in the wireless circuit breaker **200**. At block **412**, the wireless circuit breaker stores the updated fault interrupter instructions and an associated memory. Storing the updated fault interrupter instructions in the associated memory may include overwriting at least a portion of the fault interrupter instructions previously stored in the associated memory of the wireless circuit breaker **200**.

(59) At communication **414**, the wireless circuit breaker **200** communicates a validation signal to the wireless circuit breaker controller **300**. The validation signal confirms that the updated fault interrupter instructions have been validated for storage in the wireless circuit breaker **200** and/or that the updated fault interrupter instructions have been stored in the associated memory of the wireless circuit breaker **200**. At block **416**, the wireless circuit breaker **200** terminates update mode. In some examples, the wireless circuit breaker **200** can perform a self-test or other integrity check operation after the updated process, and in some cases, prior to terminating the update mode. With some examples, the wireless circuit breaker **200** may be able to restore and/or return to to pre-update condition (e.g., restore firmware from a backup, or the like) if the self-test fails. FIG. **5** illustrates a logic flow **500** associated with a wireless circuit breaker controller such as wireless circuit breaker controller **300**. The logic flow **500** may begin with block **502**. At block **502**, the wireless circuit breaker controller **300** sends an update control signal to a wireless circuit breaker (e.g., the wireless circuit breaker **200**). The update control signal is to cause the wireless circuit breaker **200** to enter an update mode. The wireless circuit breaker **200** disables tripping functionality associated therewith in the update mode. For example, the update control signal may cause the wireless circuit breaker **200** to shut down or turn off. Alternatively, the update control signal may cause the wireless circuit breaker **200** to trip or open. Furthermore, the wireless circuit breaker **200** may illuminate and associated LED (e.g., the LED **220**) in the update mode. In one implementation, the LED will flash during the update mode. For example, the LED may flash red during the update mode and then transition to a solid green after the update mode is complete.

(60) At block **504**, the wireless circuit breaker controller **300** receives an update ready signal. The update ready signal indicates that a wireless circuit breaker **200** is ready to receive updated information, such as updated fault interrupter instructions. In one embodiment, the update ready signal may be communicated to include wireless circuit breaker information that pertains to the wireless circuit breaker **200**. In one implementation, the wireless circuit breaker information includes at least a unique identifier of the wireless circuit breaker **200** and information indicating functionalities associated with the wireless circuit breaker **200**. For example, the information may identify that the wireless circuit breaker **200** is an AFCI, GFCI, or AFCI/GFCI, etc., wireless circuit breaker. The wireless circuit breaker information (e.g., the wireless circuit breaker type) may be

used by the wireless circuit breaker controller **300** to select appropriate updated fault interrupter instructions for communication to the wireless circuit breaker **200**.

(61) At block **506**, the wireless circuit breaker controller **300** communicates updated fault interrupter instructions to the wireless circuit breaker **200**.

(62) At block **508**, the wireless circuit breaker controller **300** receives a validation signal. The validation signal confirms that the updated fault interrupter instructions have been validated for storage in the wireless circuit breaker **200** and/or that the updated fault interrupter instructions have been stored in the associated memory of the wireless circuit breaker **200**.

(63) FIG. **6** illustrates a logic flow **600** associated with a wireless circuit breaker **200**. The logic flow **600** may begin with block **602**. At block **602**, a wireless circuit breaker **200** receives an update control signal or update command. The update control signal is to cause the wireless circuit breaker **200** to enter an update mode. In the update mode, the wireless circuit breaker **200** disables tripping functionality associated therewith. For example, the update control signal may cause the wireless circuit breaker **200** to shut down or turn off. Alternatively, the update control signal may cause the wireless circuit breaker **200** to trip or open. Furthermore, the wireless circuit breaker **200** may illuminate an indicator (e.g., the LED **220**) in the update mode. In one implementation, the LED will flash during the update mode. For example, the LED may flash red during the update mode and then transition to a solid green after the update mode is complete.

(64) At block **604**, the wireless circuit breaker **200** enters update mode. For example, the wireless circuit breaker **200** can open to decouple the load side phase connection from the line side phase connection. At block **606**, the wireless circuit breaker **200** determines whether the update mode was entered. For example, the wireless circuit breaker **200** can determine whether the fault interrupter is open or closed. From block **606**, the logic flow **600** can branch based on whether the update mode was entered. In particular, as depicted, logic flow **600** can continue to block **608** based on a determination that the update was entered.

(65) At block **608**, the wireless circuit breaker **200** transmits an update ready signal. The update ready signal indicates that a wireless circuit breaker **200** is ready to receive updated information, such as updated fault interrupter instructions. In one embodiment, the update ready signal may be communicated to include wireless circuit breaker information that pertains to the wireless circuit breaker **200**. In one implementation, the wireless circuit breaker information includes at least a unique identifier of the wireless circuit breaker **200** and information indicating functionalities associated with the wireless circuit breaker **200**. For example, the information may identify that the wireless circuit breaker **200** is an AFCI, GFCI, or AFCI/GFCI, etc., wireless circuit breaker. The wireless circuit breaker information (e.g., the wireless circuit breaker type) may be used by the wireless circuit breaker controller **300** to select appropriate updated fault interrupter instructions for communication to the wireless circuit breaker **200**.

(66) At block **610**, the wireless circuit breaker **200** receives updated fault interrupter instructions. In one implementation, the updated fault interrupter instructions are received from a wireless circuit breaker controller **300**. At block **612**, the wireless circuit breaker **200** transmits a validation signal. The validation signal confirms that the updated fault interrupter instructions have been validated for storage in the wireless circuit breaker and/or that the updated fault interrupter instructions have been stored in the associated memory of the wireless circuit breaker **200**. At block **614**, the wireless circuit breaker **200** terminates update mode.

(67) As detailed, from block **606**, the logic flow **600** can branch based on whether the update mode was entered. In particular, as depicted, logic flow **600** can continue to block **616** based on a determination that the update was not entered. At block **616**, the wireless circuit breaker **200** transmits an update not ready signal.

(68) FIG. **7** illustrates an embodiment of a storage medium **700**. The storage medium **700** may comprise an article of manufacture. In some examples, the storage medium **700** may include any non-transitory computer readable medium or machine readable medium, such as an optical,

magnetic or semiconductor storage. Examples of a computer readable or machine-readable storage medium may include any tangible media capable of storing electronic data, including volatile memory or non-volatile memory, removable or non-removable memory, erasable or non-erasable memory, writeable or re-writeable memory, and so forth. Examples of computer executable instructions may include any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, object-oriented code, visual code, and the like. The examples are not limited in this context. The memory **208** may be one or more memory chips capable of storing data and allowing any storage location to be directly accessed by the processor **212**, such as any type or variant of static random-access memory (SRAM), dynamic random access memory (DRAM), electrically erasable programmable read-only memory (EEPROM), Ferroelectric RAM (FRAM), NAND Flash, NOR Flash and Solid State Drives (SSD).

(69) The storage medium **700** may store various types of processor executable instructions e.g., **702**). For example, storage medium **700** can be coupled to processor(s) described herein (e.g., processor **212**, processor **222**, processor **306**, processor **314-1**, processor **314-2**, etc.) while such processor(s) can be arranged to execute instructions **702**. Thus, the storage medium **700** may store various types of computer executable instructions to implement logic flow **400**. The storage medium **700** may store various types of computer executable instructions to implement logic flow **500**. The storage medium **700** may store various types of computer executable instructions to implement logic flow **600**.

(70) Examples of a computer readable or machine-readable storage medium may include any tangible media capable of storing electronic data, including volatile memory or non-volatile memory, removable or non-removable memory, erasable or non-erasable memory, writeable or re-writeable memory, and so forth. Examples of computer executable instructions may include any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, object-oriented code, visual code, and the like. The examples are not limited in this context.

(71) While a wireless circuit breaker and panel system, a wireless circuit breaker controller, wireless technology enabled circuit breakers and a method for using the same have been described with reference to certain embodiments, it will be understood by those skilled in the art that various changes may be made, and equivalents may be substituted without departing from the spirit and scope of the claims of the application. Other modifications may be made to adapt a particular situation or material to the teachings disclosed above without departing from the scope of the claims. Therefore, the claims should not be construed as being limited to any one of the particular embodiments disclosed, but to any embodiments that fall within the scope of the claims.

(72) Furthermore, the following examples are provided to more fully described the embodiments of the present disclosure: Example 1. A communicating circuit breaker for use in a panel system with a communicating circuit breaker controller, the communicating circuit breaker comprising: a line side phase connection; a load side phase connection; a memory comprising fault interrupter instructions; a circuit interrupter; a processor communicatively coupled to the memory and the circuit interrupter; and a wireless radio configured to communicate with the communicating circuit breaker controller, wherein the processor executes the fault interrupter instructions to cause the circuit interrupter to interrupt a current flow between the line side phase connection and the load side phase connection upon detection of a fault. Example 2. The communicating circuit breaker of example 1, further comprising a power supply coupled to the line side phase connection. Example 3. The communicating circuit breaker of example 1, further comprising a miniature circuit breaker (MCB) housing. Example 4. The communicating circuit breaker of example 3, wherein the MCB housing has a width which does not exceed 1 inch. Example 5. The communicating circuit breaker of example 1, wherein the communicating circuit breaker is configured to broadcast information related to the fault detection. Example 6. The communicating circuit breaker of example 5, wherein the communicating circuit breaker is configured to broadcast the information automatically upon

detection of the fault or when instructed by the communicating circuit breaker controller. Example 7. The communicating circuit breaker of example 5, wherein the broadcast information includes one or more of a unique identifier of the communicating circuit breaker, a time stamp, and a date stamp. Example 8. The communicating circuit breaker of example 1, further comprising radio frequency (RF) shielding material disposed at least partially around the communicating circuit interrupter. Example 9. The communicating circuit breaker of example 1, wherein the wireless radio is configured to communicate using over a personal area network. Example 10. The communicating circuit breaker of example 1, the memory further comprising a firmware and firmware update instructions, wherein the communicating circuit breaker is configured to receive via the wireless radio a firmware update command from the circuit breaker controller. Example 11. The communicating circuit breaker of example 10, wherein the processor is configured to execute the firmware update instructions to cause the communicating circuit breaker to enter an update mode, the update mode comprising: the current flow between the line side phase connection and the load side phase connection being interrupted; preventing restoring of the current flow between the line side phase connection and the load side phase connection; and broadcasting a firmware update ready signal to the circuit breaker controller. Example 12. The communicating circuit breaker of example 11, wherein the current flow is interrupted by the processor sending an interrupt control signal to the circuit interrupter upon executing the firmware update instructions. Example 13. The communicating circuit breaker of example 12, wherein the current flow is prevented from being restored by the processor sending a reset disable control signal to the circuit interrupter upon executing the firmware update instructions. Example 14. The communicating circuit breaker of example 11, wherein the current flow is interrupted by a manual operation of a user. Example 15. The communicating circuit breaker of example 11, wherein the firmware update ready signal is not broadcast to the circuit breaker controller until both the current flow is interrupted and the current flow is prevented from being restored. Example 16. The communicating circuit breaker of example 11, wherein the processor is configured to execute the firmware update instructions in the update mode to further cause the communicating circuit breaker to: receive an updated firmware from the circuit breaker controller via the wireless radio; and overwrite at least a portion of the firmware in the memory with the updated firmware. Example 17. The communicating circuit breaker of example 16, wherein the processor is configured to execute the firmware update instructions in the update mode to further cause the communicating circuit breaker to: validate the updated firmware before overwriting the firmware in memory, the validation being based at least in part on a checksum validation. Example 18. The communicating circuit breaker of example 10, wherein the processor is configured to execute the firmware update instructions to cause the communicating circuit breaker to send an information element via the wireless radio to the communicating circuit breaker controller comprising an indication of at least one of a circuit breaker type, a serial number, a model number, or a firmware version number. Example 19. The communicating circuit breaker of example 1, wherein the communicating circuit breaker initiates communication via the wireless radio with the communicating circuit breaker controller automatically when the circuit interrupter interrupts the current flow between the line side phase connection and the load side phase connection. Example 20. The communicating circuit breaker of example 1, wherein the communicating circuit breaker initiates a communication via the wireless radio with the communicating circuit breaker controller automatically upon detection of the fault. Example 21. The communicating circuit breaker of example 20, wherein the communication includes fault information, the fault information including a unique identifier of the circuit breaker and a time and date of the fault detection. Example 22. The communicating circuit breaker of example 10, wherein the firmware comprises the fault interrupter instructions. Example 23. A communication enabled circuit breaker and panel system, comprising: a circuit breaker controller, comprising a first wireless radio; and at least one communication enabled circuit breaker, comprising: a line side phase connection; a load side phase connection; a memory comprising fault interrupter instructions;

a circuit interrupter; a processor communicatively coupled to the memory and the circuit interrupter; and a second wireless radio configured to communicate with the first wireless radio, wherein the processor to execute the fault interrupter instructions to cause the circuit interrupter to interrupt a current flow between the line side phase connection and the load side phase connection upon detection of a fault. Example 24. The communication enabled circuit breaker and panel system of example 23, the at least one communication enabled circuit breaker comprising, a power supply coupled to the line side phase connection. Example 25. The communication enabled circuit breaker and panel system of example 23, comprising a miniature circuit breaker (MCB) housing, the at least one communicating circuit breaker disposed in the MCB housing. Example 26. The communication enabled circuit breaker and panel system of example 25, wherein the MCB housing has a width which does not exceed 1 inch. Example 27. The communication enabled circuit breaker and panel system of example 23, comprising a plurality of communication enabled circuit breakers, each of the plurality of communication enabled circuit breakers configured to broadcast information related to the fault detection. Example 28. The communication enabled circuit breaker and panel system of example 27, wherein each of the plurality of communication enabled circuit breakers are configured to broadcast the information automatically upon detection of the fault or when instructed by the circuit breaker controller. Example 29. The communication enabled circuit breaker and panel system of example 27, wherein the broadcast information includes one or more of a unique identifier of the communication enabled circuit breaker, a time stamp, or a date stamp. Example 30. The communication enabled circuit breaker and panel system of example 23, the at least one communication enabled circuit breaker comprising radio frequency (RF) shielding material disposed at least partially around the circuit interrupter. Example 31. The communication enabled circuit breaker and panel system of example 23, wherein the second wireless radio is a personal area network wireless radio. Example 32. The communication enabled circuit breaker and panel system of example 23, the memory comprising a firmware and firmware update instructions, the at least one communication enabled circuit breaker to receive a command from the circuit breaker controller via a wireless communication channel established between the first wireless radio and the second wireless radio, the command to include an indication to update the firmware. Example 33. The communication enabled circuit breaker and panel system of example 32, the processor to execute the firmware update instructions to cause the at least one communication enabled circuit breaker to enter an update mode, comprising: sending a control signal to the circuit interrupter to cause the circuit interrupter to interrupt a current flow between the line side phase connection and the load side phase; sending a control signal to the circuit interrupter to prevent restoring the current flow between the line side phase connection and the load side phase connection; and broadcast a firmware update ready signal to the circuit breaker controller in response to the at least one communication enabled circuit breaker entering the update mode. Example 34. The communication enabled circuit breaker and panel system of example 33, the processor to execute the firmware update instructions to further cause the at least one communication enabled circuit breaker to: receive an updated firmware from the circuit breaker controller via the wireless communication channel; and overwrite at least a portion of the firmware in the memory with the updated firmware based on the at least one communication enabled circuit breaker entering the update mode. Example 35. The communication enabled circuit breaker and panel system of example 34, the processor to execute the firmware update instructions to further cause the at least one communication enabled circuit breaker to: validate an of the updated firmware received from the circuit breaker controller based at least in part on a checksum validation; and overwrite at least a portion of the firmware in the memory with the updated firmware in response to the updated firmware passing the checksum validation. Example 36. The communication enabled circuit breaker and panel system of example 33, the processor to execute the firmware update instructions to further cause the at least one communication enabled circuit breaker to send an information element to the circuit breaker controller comprising an indication of at least one of a circuit breaker type, a serial number, a model number, or a firmware version number. Example 37. The communication enabled circuit breaker and panel system of example 23, wherein the first wireless radio initiates wireless communication with the second wireless radio automatically when the circuit interrupter interrupts the current flow between the line side phase connection and the load side phase connection. Example 38. The communication enabled circuit breaker and panel system of example 37, the wireless communication includes at least fault information including a unique identifier of the circuit breaker and a time and date of a trip incident that caused the fault interrupter to interrupt the current flow between the line side phase connection and the load side phase connection. Example 39. The communication enabled circuit breaker and panel system of example 32, wherein the firmware comprises the fault interrupter instructions. Example 40. A method to update a wireless circuit breaker, comprising: wirelessly receiving an update control signal at the wireless circuit breaker; initiating an update mode of the wireless circuit breaker; wirelessly receiving updated fault interrupter instructions at the wireless circuit breaker; and validating the wirelessly received updated fault interrupter instructions. Example 41. The method of example 40, wherein the updated control signal is received from a circuit breaker controller, a mobile phone, or the Internet. Example 42. The method of example 40, comprising: updating the wireless circuit breaker based in part on the updated fault interrupter instructions; and running a self-test on the wireless circuit breaker in response to the update. Example 43. The method of example 42, comprising restoring the wireless circuit breaker to a pre-update condition in response to a failure of the self-test. Example 44. The method of example 40, comprising overwriting at least a portion of stored fault interrupter instructions with the updated fault interrupter instructions. Example 45. The method of example 40, wherein initiating the update mode of the wireless circuit breaker comprises at least disabling resetting functionality of the wireless circuit breaker. Example 46. The method of example 45, wherein initiating the update mode of the wireless circuit breaker comprises turning off the circuit breaker. Example 47. The method of example 40, wherein initiating the update mode of the wireless circuit breaker comprises illuminating a light emitting diode (LED) of the wireless circuit breaker to indicate that the wireless circuit breaker is in the update mode. Example 48. A wireless circuit breaker and panel system, comprising: a wireless circuit breaker controller, comprising a first wireless radio; and at least one wireless circuit breaker, comprising: a line side phase connection; a memory comprising fault interrupter instructions; a fault interrupter coupled to the memory, the fault interrupter to interrupt a current flow on the line side phase connection based at least in part on the fault interrupter instructions; and a second wireless radio coupled to the memory, the second wireless radio to receive a signal from the first wireless radio, the signal to include an indication to update the fault interrupter instructions. Example 49. The wireless circuit breaker and panel system of example 48, the wireless circuit breaker comprising a load side phase connection, a power supply coupled to the line side phase connection, the memory, and the second wireless radio, the power supply configured to provide power to the memory and the second wireless radio, and the fault interrupter to interrupt a current flow between the line side phase connection and the load side phase connection based at least in part on the fault interrupter instructions. Example 50. The wireless circuit breaker and panel system of example 48, comprising a miniature circuit breaker (MCB) housing, the at least one wireless circuit breaker disposed in the MCB housing. Example 51. The wireless circuit breaker and panel system of example 50, the MCB housing has a width which does not exceed 1 inch. Example 52. The wireless circuit breaker and panel system of example 48, comprising a plurality of wireless circuit breakers, each of the plurality of wireless circuit breakers comprising a second wireless radio and a processor, the processor of each of the plurality of wireless breakers configured to initiate wireless communication of the second wireless radio only in the event of a trip incident or when instructed by the wireless circuit breaker controller. Example 53. The wireless circuit breaker and panel system of example 48, comprising a radio frequency (RF) shielding material at least partially surrounding the fault interrupter, the RF shielding material to attenuate wireless communication signals. Example 54. The wireless circuit breaker and panel system of example 53, the wireless communication includes at least fault information including a unique identifier of the at least one wireless circuit breaker and a time and date of a trip incident that caused the fault interrupter to interrupt the current flow on the line side phase connection. Example 55. The wireless circuit breaker and panel system of example 48, the second wireless radio initiates wireless communication with the wireless circuit breaker controller automatically when the fault interrupter interrupts the current flow on the line side phase connection. Example 56. The wireless circuit breaker and panel system of example 48, the second wireless radio is a Bluetooth® enabled wireless radio. Example 57. A wirelessly upgradable wireless circuit breaker, comprising: a line side phase connection; a load side phase connection; a memory comprising fault interrupter instructions; a fault interrupter coupled to the memory, the fault interrupter to interrupt a current flow between the line side phase connection and the load side phase connection based at least in part on the fault interrupter instructions; and a wireless radio coupled to the memory, the wireless radio to receive a wireless signal, the wireless signal to include an indication to update the fault interrupter instructions. Example 58. The wirelessly upgradable wireless circuit breaker of example 57, comprising a power supply coupled to the line side phase connection, the memory, and the wireless radio, the power supply configured to source power from the line side phase connection and supply power to the memory and the wireless radio. Example 59. The wirelessly upgradable wireless circuit breaker of example 57, comprising a miniature circuit breaker (MCB) housing, the wirelessly upgradable wireless circuit breaker disposed in the MCB housing. Example 60. The wirelessly upgradable wireless circuit breaker of example 59, the MCB housing has a width of 1 inch. Example 61. The wirelessly upgradable wireless circuit breaker of example 57, comprising a radio frequency (RF) shielding material at least partially surrounding the fault interrupter, the RF shielding material to attenuate wireless communication signals. Example 62. The wirelessly upgradable wireless circuit breaker of example 57, the wireless radio initiates wireless communication automatically when the fault interrupter interrupts the current flow between the line side phase connection and the load side phase connection. Example 63. The wirelessly upgradable wireless circuit breaker of example 62, the wireless communication includes at least fault information including a unique identifier of the wirelessly upgradable wireless circuit breaker and a time and date of a trip incident that caused the fault interrupter to interrupt the current flow between the line side phase connection and the load side phase connection. Example 64. The wirelessly upgradable wireless circuit breaker of example 57, the wireless radio is a Bluetooth® enabled wireless radio. Example 65. A method to update a wireless circuit breaker, comprising: wirelessly receiving an update control signal at the wireless circuit breaker; initiating an update mode of the wireless circuit breaker; wirelessly receiving updated fault interrupter instructions at the wireless circuit breaker; and validating the wirelessly received updated fault interrupter instructions. Example 66. The method to update the wireless circuit breaker of example 65, comprising overwriting at least a portion of stored fault interrupter instructions with the updated fault interrupter instructions. Example 67. The method to update the wireless circuit breaker of example 65, the initiating the update mode of the wireless circuit breaker includes at least disabling tripping functionality of the wireless circuit breaker. Example 68. The method to update the wireless circuit breaker of example 65, the initiating the update mode of the wireless circuit breaker includes illuminating a light emitting diode (LED) of the wireless circuit breaker to indicate that the wireless circuit breaker is in the update mode. Example 69. A method to provide updated fault interrupter instructions, comprising: wirelessly receiving an update ready signal indicating that a wireless circuit breaker is in an update mode; wirelessly transmitting updated fault interrupter instructions, the updated fault interrupter instructions wirelessly transmitted by a wireless circuit breaker controller; and receiving a validation signal indicating that the updated fault interrupter instructions were received by the wireless circuit breaker. Example 70. The method to provide the updated fault interrupter instructions of example 69, the update ready signal indicates that the wireless circuit breaker has disabled tripping functionality of the wireless circuit breaker. Example 71. The method to provide the updated fault interrupter instructions of example 69, the validation signal confirms that the updated fault interrupter instructions are stored in a memory of the wireless circuit breaker. Example 72. The method to provide the updated fault interrupter instructions of example 69, comprising wirelessly transmitting an update control signal to the wireless circuit breaker, and wirelessly receiving the update ready signal in response to the update control signal, the update ready signal including wireless circuit breaker information. Example 73. The method to provide the updated fault interrupter instructions of example 72, the wireless circuit breaker information includes at least a unique identifier of the wireless circuit breaker and information indicating functionalities associated with the wireless circuit breaker.

Claims

1. A communicating circuit breaker for use in a panel system, the panel system having a plurality of communicating circuit breakers in a panel, the communicating circuit breaker comprising: a miniature circuit breaker (MCB) housing comprising: a memory to comprise circuit breaker instructions; a processor coupled with the memory to execute the circuit breaker instructions; a first wireless radio configured to establish a first direct wireless connection with a wireless circuit breaker controller in the panel; and a second wireless radio configured to establish a direct wireless connection with a mobile device that is separate from the panel system, wherein the mobile device is located outside of the panel of the panel system; wherein the communicating circuit breaker is configured to validate new circuit breaker instructions received from: the wireless circuit breaker controller via the first wireless radio; or the mobile device via the second wireless radio.
2. The communicating circuit breaker of claim 1 wherein the circuit breaker instructions comprise one or more selected from the group consisting of operating instructions or fault interrupter instructions.
3. The communicating circuit breaker of claim 1, wherein the second wireless radio is configured to communicate over a network with a remote entity.
4. The communicating circuit breaker of claim 1, wherein the second wireless radio comprises at least one of a personal area network radio, a cellular radio, another network radio, or a combination thereof.
5. The communicating circuit breaker of claim 1, wherein the second wireless radio is coupled with an antenna.
6. The communicating circuit breaker of claim 5, wherein the second wireless radio is configured to communicate with the mobile device via the antenna to: receive new circuit breaker instructions or an update for the circuit breaker instructions from the mobile device; receive other updated instructions; receive operating software; receive firmware; configure the communicating circuit breaker; report status information; transfer historical data; transfer diagnostic information; or a combination thereof.
7. The communicating circuit breaker of claim 1, wherein the wireless radios are configured to: wirelessly receive an update control signal at the communicating circuit breaker; initiate an update mode of the communicating circuit breaker; wirelessly receive updated circuit breaker instructions at the communicating circuit breaker; validate the wirelessly received updated circuit breaker instructions; and update the communicating circuit breaker with the wirelessly received updated circuit breaker instructions.
8. The communicating circuit breaker of claim 7, further comprising a fault interrupter, the processor to update the communicating circuit breaker with the wirelessly received updated circuit breaker instructions, the update configured to occur while the fault interrupter is in a tripped state.
9. The communicating circuit breaker of claim 8, the processor to reset a fault interrupter after updating the communicating circuit breaker with the wirelessly received updated circuit breaker

instructions.
10. The communicating circuit breaker of claim 7, wherein the fault interrupter remains set while updating the communicating circuit breaker with the wirelessly received updated circuit breaker instructions.
11. A method for a communicating circuit breaker, for use with a communicating circuit breaker controller and a mobile device, within a panel of a panel system, the method comprising the steps of: executing, by a processor in a miniature circuit breaker (MCB) housing of the communicating circuit breaker, circuit breaker instructions to cause a circuit interrupter in the communicating circuit breaker to interrupt a current flow between a line side phase connection and a load side phase connection of the communicating circuit breaker upon detection of a fault; establishing a first direct wireless connection via a first wireless radio with a communicating circuit breaker controller in the panel; establishing a second direct wireless connection via a second wireless radio with a mobile device that is separate from the panel system and wherein the mobile device is located outside of the panel; and selectively transmitting information on the first and second direct wireless connections, wherein the information comprises one or more of a status, a trip alarm, fault information, or operating parameters; wherein the communicating circuit breaker is configured to validate new circuit breaker instructions received from: the communicating circuit breaker controller via the first wireless radio; or the mobile device via the second wireless radio.
12. The method of claim 11, wherein establishing one or more of the first direct wireless connection or establishing the second direct wireless connection comprises establishing at least one of a personal area network radio, a cellular radio, an other network radio, or a combination thereof.
13. The method of claim 11, further comprises communicating via an antenna coupled with the second wireless radio of the communicating circuit breaker to: receive new circuit breaker instructions or an update for the circuit breaker instructions from the mobile device via the antenna; receive other updated instructions; receive operating software; receive firmware; configure the communicating circuit breaker, to report status information; , te-transfer historical data; transfer diagnostic information; or a combination thereof.
14. The method of claim 11, further comprising: wirelessly receiving an update control signal at the communicating circuit breaker; initiating an update mode of the communicating circuit breaker; wirelessly receiving updated circuit breaker instructions at the communicating circuit breaker; validating the wirelessly received updated circuit breaker instructions; and updating the communicating circuit breaker with the wirelessly received updated circuit breaker instructions.
15. The method of claim 11, wherein the step of updating the communicating circuit breaker with the wirelessly received updated circuit breaker instructions further comprises updating only a portion of the circuit breaker instructions.
16. The method of claim 15, wherein while in the update mode: the communicating circuit breaker remains set; and upon detection of a fault, the current flow between a line side phase connection and the load side phase connection is interrupted.
17. The method of claim 14, further comprising resetting a fault interrupter after updating the communicating circuit breaker with the wirelessly received updated circuit breaker instructions.
18. The method of claim 14, wherein a fault interrupter of the communicating circuit breaker remains set while updating the communicating circuit breaker with the wirelessly received updated circuit breaker instructions.
19. A machine-readable, non-transitory storage medium comprising instructions, which, when executed by a processor, causes the processor to perform operations to: execute circuit breaker instructions to cause a fault interrupter of a communicating circuit breaker to interrupt a current flow between a line side phase connection and a load side phase connection of the communicating circuit breaker upon detection of a fault; establish a first direct wireless connection via a first wireless radio with a communicating circuit breaker controller within a panel of a panel system, wherein the processor is within a miniature circuit breaker (MCB) housing of the communicating

circuit breaker within the panel of the panel system; establish a second direct wireless connection via a second wireless radio with a mobile device that is separate from the panel system and wherein the mobile device is located outside of the panel; and selectively transmit information on the first and second direct wireless connections, wherein the information comprises one or more of a status, a trip alarm, fault information, or operating parameters for the communicating circuit breaker; wherein the communicating circuit breaker is configured to validate new circuit breaker instructions received from: the communicating circuit breaker controller via the first wireless radio; or the mobile device via the second wireless radio.

20. The storage medium of claim 19, wherein the operations to establish the first direct wireless connection and establish the second direct wireless connection comprises operations to establish at least one of a personal area network radio, a cellular radio, another network radio, or a combination thereof.

21. The storage medium of claim 19, wherein the operations further comprise operations to communicate via an antenna coupled with the second wireless radio, to receive new circuit breaker instructions or an update for the circuit breaker instructions from the mobile device, to receive other updated instructions, to receive operating software, to receive firmware, to configure the communicating circuit breaker, to report status information, to transfer historical data, to transfer diagnostic information, or a combination thereof.

22. The storage medium of claim 19, wherein the operations further comprise operations to: wirelessly receive an update control signal at the communicating circuit breaker; initiate an update mode of the communicating circuit breaker; wirelessly receive updated circuit breaker instructions at the communicating circuit breaker; validate the wirelessly received updated circuit breaker instructions; and update the communicating circuit breaker with the wirelessly received updated circuit breaker instructions.

23. The storage medium of claim 22, wherein the operations further comprise operations to reset the fault interrupter after updating the communicating circuit breaker with the wirelessly received updated circuit breaker instructions.

24. The storage medium of claim 22, wherein the fault interrupter remains set while updating the communicating circuit breaker with the wirelessly received updated circuit breaker instructions.

25. The communicating circuit breaker of claim 1, wherein: the communicating circuit breaker is a single-pole circuit breaker and the MCB housing has a width which does not exceed 1 inch; or the communicating circuit breaker is a two-pole circuit breaker and the MCB housing has a width which does not exceed 2 inches.